

Accurate QM/MM Molecular Dynamics for Periodic Systems in GPU4PySCF with Applications to Enzyme Catalysis

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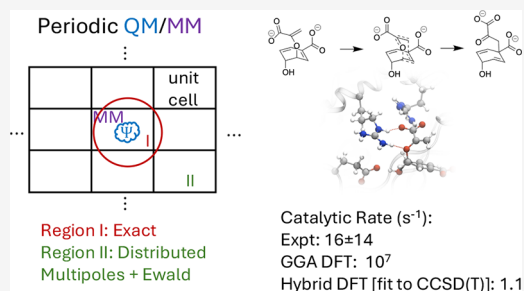


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ABSTRACT: We present an implementation of the quantum mechanics/molecular mechanics (QM/MM) method for periodic systems using GPU accelerated QM methods, a distributed multipole formulation of the electrostatics, and a pseudobond treatment of the QM/MM boundary. We demonstrate that our method has well-controlled errors, stable self-consistent QM convergence, and energy-conserving dynamics. We further describe an application to the catalytic kinetics of chorismate mutase. Using an accurate hybrid functional reparametrized to coupled cluster energetics, our QM/MM simulations highlight the sensitivity in the calculated rate to the choice of quantum method, quantum region selection, and local protein conformation. Our work is provided through the open-source PySCF package using acceleration from the GPU4PySCF module.



INTRODUCTION

Quantum mechanics (QM) is the underlying physical theory of chemical phenomena. However, applying a full QM treatment to large-scale chemical problems is both unnecessary and computationally prohibitive. Pioneered by Warshel, Levitt, and Karplus,^{1,2} the hybrid Quantum Mechanics/Molecular Mechanics (QM/MM) approach addresses this by limiting the QM treatment to critical regions of interest while modeling the surroundings with more approximate empirical force fields.^{3–10}

In this work, we describe and apply a new QM/MM implementation. Given a QM/MM system partitioning, the factors that define the QM/MM treatment are (i) the choice of method to use in the QM calculation, (ii) the treatment of the electrostatic interaction between the QM and MM regions, (iii) the handling of the QM/MM boundary, and (iv) the quality of the MM force-field. We present a GPU-based QM/MM implementation with the following characteristics (i) fast algorithms for accurate QM methods, such as hybrid density functionals, in the QM region, (ii) a multipole representation and self-consistent treatment of the periodic quantum electrostatic potential (denoted here the QM/MM-Multipole method), (iii) a pseudobond treatment of the QM/MM boundary, and an open-source implementation within PySCF taking advantage of the GPU accelerated GPU4PySCF module,^{11,12} thus (iv) enabling straightforward integration with standard open-source force-fields.

We carefully benchmark the convergence of the QM/MM-Multipole method (and its resulting effect on the QM self-consistent field stability and energy conservation) in a QM/MM water model and a model for microtubule-mediated GTP hydrolysis. Finally, we investigate the utility of our implementation in enzyme reactions in the context of the

catalytic kinetics of chorismate mutase, which has recently been studied at the QM/MM level using a pure density functional description of the QM region.¹³ In our simulations, we are able to employ a more accurate QM method obtained by reparametrizing a hybrid density functional to local coupled cluster energetics. Our resulting reaction rate, extracted by enhanced sampling of the QM/MM potential energy surface, improves the agreement with the experimental estimate, and overall our study highlights the sensitivity of the rate to many aspects of the QM/MM treatment.

THEORY AND METHODS

Energy Partitioning. We consider a total system with periodic boundary conditions consisting of two subsystems, partitioned as two sets of atoms. One subsystem is then described by quantum chemistry methods (in this work, some form of density functional theory) while the other subsystem is described by empirical classical force fields. For the definition of the total energy, we use the electrostatic embedding scheme, where the QM subsystem interacts with nonpolarizable MM atoms via a classical Coulomb interaction. Specifically, the system energy functional is the sum of three components,

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$$E(\Psi, \mathbf{R}^{\text{QM}}, \mathbf{R}^{\text{MM}}) = E_{\text{QM}}(\Psi, \mathbf{R}^{\text{QM}}) + E_{\text{QM-MM}}(\rho(\mathbf{r}), \mathbf{R}^{\text{MM}}) + E_{\text{MM}}(\mathbf{R}^{\text{QM}}, \mathbf{R}^{\text{MM}}) \quad (1)$$

where E_{QM} is the energy functional of the (periodic) QM subsystem without the MM field (i.e., it is a function of the electronic wave function Ψ and the QM atom positions $\mathbf{R}^{\text{QM}}$), E_{MM} is the energy of the (periodically repeated) MM atoms themselves plus the van der Waals interactions between the QM and MM atoms, both of which are described at the MM level, and $E_{\text{QM-MM}}$ denotes the electrostatic coupling between the QM electron density $\rho(\mathbf{r})$ and the charges centered on the MM atoms. We minimize this functional with respect to Ψ , which relaxes the QM density in the presence of the MM charges and produces a self-consistent treatment of the electrostatic QM/MM coupling. We then obtain the energy and gradients in order to carry out ab initio QM/MM molecular dynamics (MD).

Electrostatic Treatment: Background. In condensed phase systems, a critical component of the functional in eq 1 is the electrostatic coupling, in particular the contributions that arise from long-range periodic electrostatics within each of the three terms. It has long been recognized that serious artifacts arise in molecular simulations from ignoring the long-range tail of the electrostatics.^{14,15}

Long-range electrostatics E_{MM} are part of standard classical MD methodology and are handled in the force-field part of the QM/MM implementation thus we do not discuss them further here. The electrostatic contributions from the QM charge density in the first two terms of eq 1, however, require a different treatment, due to the continuous nature of the QM charge density. Depending on the formulation, the computation of electrostatics involving this charge density can be quite expensive. To motivate our formulation, we briefly summarize some background on how other QM/MM implementations treat the periodic electrostatic coupling between the QM density and MM charges, before discussing our formulation in the next section. (Because we will treat the QM-QM electrostatic interaction in the same way as the QM-MM electrostatic interaction, we will not separately survey the QM-QM electrostatic treatments in the literature).

One category of methods represents the electrostatic potential of the periodic MM charge distribution on a grid constructed within the reference cell. The MM electrostatic potential can then be integrated with the QM charge density in the reference cell.^{16,17} Various kinds of grids and approximations to accelerate this computation have been explored.^{16,18–22}

An alternative approach, named the QM/MM-Ewald method and originally proposed in the context of semi-empirical QM treatments,^{23–31} approximates the QM electron distribution by QM atom-centered point charges. The long-range QM-MM electrostatics, i.e. the interaction between the reference cell QM density and periodic MM charges outside of the reference cell, can then be handled using standard point-charge Ewald lattice sum methods, while the interactions between the QM density and nearby MM charges within the reference cell are computed exactly.

The QM/MM-Ewald method has the benefit of simplicity and avoids the costly evaluation of the long-range electrostatic potential on a large number of grid points, but it assumes that the QM charge distribution is well approximated by point charges. This is a good approximation when there is a large

separation between the reference cell QM charges and the MM charges in other cells from a large simulation box. However, artifacts have been reported when applying this together with a DFT treatment of the QM region. In particular, difficulties were observed in converging the self-consistent field (SCF) iterations when using Mulliken QM charges and a diffuse basis,^{27,32} and a stable SCF required refitting the QM charges to the QM electrostatic potential during the SCF cycles.^{27,30}

Multipole Approximation to Electrostatics. Our approach to the long-range electrostatics starts from the simple and efficient QM/MM-Ewald method but extends the QM electronic description to a multipole representation.^{33–45} As we show later in the numerical results, this yields favorable convergence and SCF stability properties without the need to refit the QM charges during the SCF cycles.^{27,30}

To start, we partition the periodic QM/MM problem into two regions (Figure 1). Region I contains the QM subsystem

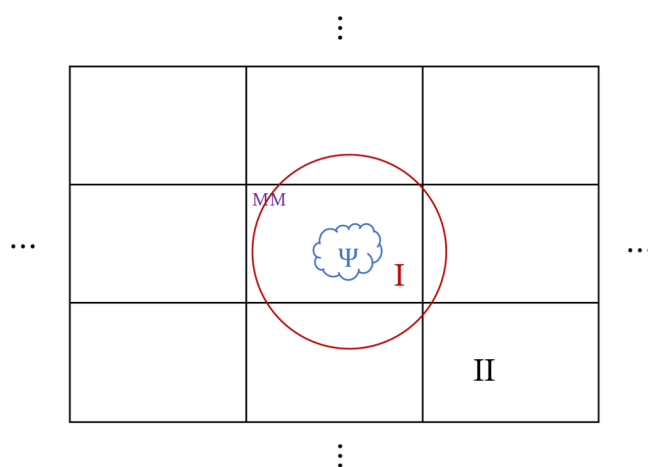


Figure 1. Illustration of the partitioning of the QM-MM electrostatics into near-field (region I) and far-field (region II) components. Each rectangle represents one unit cell. Region I is shown as a sphere centered on the QM subsystem and region II contains the remaining infinite lattice.

in the reference cell and the charges in the MM region that are close to the reference cell. Region II, the distant area, includes infinitely many atoms (i.e., the entire periodic lattice) and is where controlled approximations are made. It is assumed that the QM periodic images are always in region II, a convention maintained in our implementation and subsequent discussions. The electrostatic components of the energies E_{QM} and $E_{\text{QM-MM}}$ are split into contributions from these two regions, labeled E^{I} and E^{II} .

The QM-MM interaction within region I is computed using the exact QM density:

$$E_{\text{QM-MM}}^{\text{I}} = \sum_{i \in \text{I}} \int d\mathbf{r} \frac{-\rho(\mathbf{r})q_i}{|\mathbf{r} - \mathbf{R}_i^{\text{MM}}|} = \sum_{i \in \text{I}} \sum_{\mu\nu} \int d\mathbf{r} \frac{-\phi_{\mu}(\mathbf{r})\phi_{\nu}(\mathbf{r})q_i}{|\mathbf{r} - \mathbf{R}_i^{\text{MM}}|} \gamma_{\mu\nu} \quad (2)$$

where we assume the MM charges are point charges q_i at the MM atom positions, and we use $\gamma_{\mu\nu}$ to denote the one-body reduced density matrix element (1-RDM) in the QM region, and $\{\phi_{\mu}(\mathbf{r})\}$ to represent the atom-centered computational

basis functions. (The extension to Gaussian-distributed MM charges is discussed later).

For the QM-MM coupling in region II, we approximate the electrostatic potential around QM basis center $\mathbf{R}_\mu^{\text{QM}}$, arising from an MM charge at position $\mathbf{R}_i^{\text{MM}}$, using a multipole expansion (Taylor series) around $\mathbf{R}_\mu^{\text{QM}}$, (α and β index Cartesian components x , y , and z in below):

$$\frac{1}{|\mathbf{r} - \mathbf{R}_i^{\text{MM}}|} \approx \frac{1}{|\mathbf{R}_\mu^{\text{QM}} - \mathbf{R}_i^{\text{MM}}|} - \frac{(\mathbf{R}_\mu^{\text{QM}} - \mathbf{R}_i^{\text{MM}}) \cdot (\mathbf{r} - \mathbf{R}_\mu^{\text{QM}})}{|\mathbf{R}_\mu^{\text{QM}} - \mathbf{R}_i^{\text{MM}}|^3} + \frac{1}{2} \sum_{\alpha\beta} \frac{3(R_{\mu\alpha}^{\text{QM}} - R_{i\alpha}^{\text{MM}})(R_{\mu\beta}^{\text{QM}} - R_{i\beta}^{\text{MM}}) - \delta_{\alpha\beta}|\mathbf{R}_\mu^{\text{QM}} - \mathbf{R}_i^{\text{MM}}|^2}{|\mathbf{R}_\mu^{\text{QM}} - \mathbf{R}_i^{\text{MM}}|^5} (r_\alpha - R_{i\alpha}^{\text{MM}})(r_\beta - R_{i\beta}^{\text{MM}}) \quad (3)$$

For the QM density contribution of a basis pair μ and ν centered on different atoms, the expansion is performed at both centers and then arithmetically averaged. The Taylor series in eq 3 has been truncated at the second order. The leading error in such an approximation is the missing third-order term that scales asymptotically as $1/|\mathbf{R}_\mu^{\text{QM}} - \mathbf{R}_i^{\text{MM}}|^4$. This is the first term in the series that decays sufficiently fast to ensure its infinite lattice sum (over the periodic MM images) absolutely converges in real space.⁴⁶ This allows for control over the truncation error to achieve the desired accuracy by appropriately selecting Region I. We discuss how this can be accomplished in practice later.

It is straightforward to show that substituting the expansion (eq 3) into the QM-MM electrostatic integral leads to a distributed multipole expansion approximation centered on the QM atoms:

$$E_{\text{QM-MM}}^{\text{II}} \approx \sum_{i \in \text{MM(II)}} \sum_{j \in \text{QM(I)}} [Q_j q_i T_{ji} + \sum_{\alpha} \mu_{j\alpha} q_i T_{jia} + \sum_{\alpha\beta} \theta_{j\alpha\beta} q_i T_{jia\beta}] \quad (4)$$

where Q , μ and θ are Mulliken atomic charges, dipoles and quadrupoles,⁴⁷ and the $T_{ji\dots}$ are the Taylor expansion coefficients, referred to as the multipole interaction tensors. A similar multipole expansion approximation describes the interactions between the unit-cell QM and periodic QM images, yielding:

$$E_{\text{QM}}^{\text{II}} \approx \frac{1}{2} \sum_{i \in \text{QM(II)}} \sum_{j \in \text{QM(I)}} [Q_j Q_i T_{ji} + \sum_{\alpha} \mu_{j\alpha} Q_i T_{jia} + \sum_{\alpha\beta} \theta_{j\alpha\beta} Q_i T_{jia\beta} + 2 \sum_{\alpha\beta} \mu_{j\alpha} \mu_{i\beta} T_{jia\beta}] \quad (5)$$

We ignore the electronic exchange and correlation effects between QM images, due to their short-range nature. To manage the infinite lattice sum over region II, we utilize the identity $\sum_{i \in \text{II}} = \sum_i - \sum_{i \in \text{I}}$, converting the restricted sum over region II into an unrestricted sum over the whole lattice minus a finite real-space sum over region I. We treat the whole lattice sum by the Ewald method for multipoles⁴⁸ (see also SI), and we rewrite the QM/MM energy of the periodic system as

$$E(\Psi, \mathbf{R}^{\text{QM}}, \mathbf{R}^{\text{MM}}) = E_{\text{QM}}^{\text{iso}}(\Psi, \mathbf{R}^{\text{QM}}) + E_{\text{QM-MM}}^{\text{I}}(\rho(\mathbf{r}), \mathbf{R}^{\text{QM}}, \mathbf{R}^{\text{MM}}) - \frac{1}{2} \sum_{ij \in \text{QM(I)} \& i \neq j} [Q_j Q_i T_{ji} + \sum_{\alpha} \mu_{j\alpha} Q_i T_{jia} + \sum_{\alpha\beta} \theta_{j\alpha\beta} Q_i T_{jia\beta} + 2 \sum_{\alpha\beta} \mu_{j\alpha} \mu_{i\beta} T_{jia\beta}] + \frac{1}{2} \sum_{ij \in \text{QM(I)}} [Q_j Q_i \psi_{ji} + \sum_{\alpha} \mu_{j\alpha} Q_i \psi_{jia} + \sum_{\alpha\beta} \theta_{j\alpha\beta} Q_i \psi_{jia\beta} + 2 \sum_{\alpha\beta} \mu_{j\alpha} \mu_{i\beta} \psi_{jia\beta}] - \sum_{i \in \text{MM(I)}} \sum_{j \in \text{QM(I)}} [Q_j q_i T_{ji} + \sum_{\alpha} \mu_{j\alpha} q_i T_{jia} + \sum_{\alpha\beta} \theta_{j\alpha\beta} q_i T_{jia\beta}] + \sum_{i \in \text{MM(unitcell)}} \sum_{j \in \text{QM(I)}} [Q_j q_i \psi_{ji} + \sum_{\alpha} \mu_{j\alpha} q_i \psi_{jia} + \sum_{\alpha\beta} \theta_{j\alpha\beta} q_i \psi_{jia\beta}] + E_{\text{MM}}(\mathbf{R}^{\text{QM}}, \mathbf{R}^{\text{MM}}) \quad (6)$$

where $E_{\text{QM}}^{\text{iso}}$ indicates the QM energy of an isolated QM subsystem i.e. without the periodic QM-QM electrostatic interactions. In eq 6, the first two terms account for all the QM and QM-MM energies in the region I. The third and fourth terms sum to $E_{\text{QM}}^{\text{II}}$, and the fifth and sixth terms sum to $E_{\text{QM-MM}}^{\text{II}}$. The Ewald interaction tensors are denoted as $\psi_{ji\dots}$ and are given in the SI. (When we write a summation range over the unit cell, we are assuming the minimum image convention).

In addition to the MM point charge model, it is useful to consider Gaussian smeared MM charges. Using such Gaussian charges reduces the so-called “electron spill-out” effect,^{7,20,23} where density that is very close to a point MM charge is overpolarized. In this context, $E_{\text{QM-MM}}^{\text{I}}$ is calculated as

$$E_{\text{QM-MM}}^{\text{I}} = \sum_{i \in \text{MM(I)}} \sum_{\mu\nu} \int d\mathbf{r}_1 d\mathbf{r}_2 \frac{-\phi_\mu(\mathbf{r}_1) \phi_\nu(\mathbf{r}_1) \phi_i(\mathbf{r}_2) q_i}{|\mathbf{r}_1 - \mathbf{r}_2|} \gamma_{\mu\nu} \quad (7)$$

where ϕ_i is a normalized Gaussian function with exponent κ_i , centered on MM atom i , representing its charge distribution. The energy expression for MM point charges (eq 6) is then used with a correction term to reconcile the discrepancy between Gaussian distributed charges and point charges in their interactions with QM multipoles,

$$\Delta E_{\text{QM-MM}}^{\text{II}} = - \sum_{i \in \text{MM(II)}} \sum_{j \in \text{QM(I)}} [Q_j q_i \hat{T}_{ji} + \sum_{\alpha} \mu_{j\alpha} q_i \hat{T}_{jia} + \sum_{\alpha\beta} \theta_{j\alpha\beta} q_i \hat{T}_{jia\beta}] \quad (8)$$

where $\hat{T}_{ji\dots}$ represents the multipole interaction tensors between point charges and Gaussian distributed charges (detailed in the SI). Since this correction decays exponentially fast with the charge separation, the infinite lattice sum over MM charges in the above expression reduces to a finite sum in real space, and the truncation can be made based on each MM charge exponent κ_i .

Electronic Mean-Field Theory. The Schrödinger equation that Ψ satisfies may be derived through the variational minimization of the QM/MM energy eq 6 with respect to Ψ . Within a mean-field approximation of the electronic structure,

the wave function takes a single Slater determinant form, parametrized by the 1-RDM γ . Depending on the mean-field energy functional, the variational minimization of the QM/MM energy leads to the solution of Hartree–Fock (HF) or Kohn–Sham (KS) equations. The Fock/Kohn–Sham matrix that includes the QM/MM interactions is derived by taking the gradient of the QM/MM energy with respect to γ ,

$$f_{\mu\nu} = f_{\mu\nu}^I + \sum_{j \in \text{QM(I)}} [v_j^{\text{II}} \frac{\partial Q_j}{\partial \gamma_{\mu\nu}} + \sum_{\alpha} v_{j\alpha}^{\text{II}} \frac{\partial \mu_{j\alpha}}{\partial \gamma_{\mu\nu}} + \sum_{\alpha\beta} v_{j\alpha\beta}^{\text{II}} \frac{\partial \theta_{\alpha\beta}}{\partial \gamma_{\mu\nu}}] \quad (9)$$

In the above, $f^I = \partial(E_{\text{QM}}^{\text{iso}} + E_{\text{QM-MM}}^I)/\partial\gamma$ is the Fock operator for the nonperiodic QM/MM system consisting of the unit-cell QM atoms and the region I MM atoms. The potential v^{II} that the QM multipoles experience due to their interactions with region II is derived by differentiating the multipole-dependent terms in eq 6 (and eq 8 if using Gaussian-distributed MM charges), specifically, $(v_j^{\text{II}}, v_{j\alpha}^{\text{II}}, v_{j\alpha\beta}^{\text{II}}) = \partial E^{\text{II}}/\partial(Q_j, \mu_{j\alpha}, \theta_{j\alpha\beta})$. The response of the Mulliken multipoles to changes in the 1-RDM is given by overlap-type integrals,

$$\frac{\partial Q_j}{\partial \gamma_{\mu\nu}} = -\frac{1}{2}(\delta_{\mu \in j} + \delta_{\nu \in j})\langle \mu | \nu \rangle \quad (10)$$

$$\frac{\partial \mu_{j\alpha}}{\partial \gamma_{\mu\nu}} = -\frac{1}{2}(\delta_{\mu \in j} + \delta_{\nu \in j})\langle \mu | (r_{\alpha} - R_{j\alpha}) | \nu \rangle \quad (11)$$

$$\frac{\partial \theta_{j\alpha\beta}}{\partial \gamma_{\mu\nu}} = -\frac{1}{2}(\delta_{\mu \in j} + \delta_{\nu \in j})\langle \mu | (r_{\alpha} - R_{j\alpha})(r_{\beta} - R_{j\beta}) | \nu \rangle \quad (12)$$

where $\delta_{\mu \in j}$ signifies whether orbital μ is centered on atom j , and $\langle \mu | f(\mathbf{r}) | \nu \rangle$ is a shorthand notation for $\int d\mathbf{r} \phi_{\mu}(\mathbf{r})f(\mathbf{r})\phi_{\nu}(\mathbf{r})$. As discussed earlier, the Taylor expansion (eq 3) is conducted at both centers of orbitals μ and ν and then averaged, and thus the 1/2 factor arises. Such an averaged expansion is equivalent to symmetrizing the Fock operator resulting from a single-sided Taylor expansion centered on only one orbital.

The nuclear gradient required to perform MD is obtained by differentiating the QM/MM energy with respect to the atom positions. The nuclear gradient of the interaction energy in the region I ($E_{\text{QM}}^{\text{iso}} + E_{\text{QM-MM}}^I$) is well-documented in the literature for nonperiodic QM/MM systems (see e.g., ref 3). The E^{II} gradient contains both the response of the multipole interaction tensors $T_{j,\dots}$, $\psi_{j,\dots}$, and $\hat{T}_{j,\dots}$ and of the multipole moments Q_j , μ and θ . The interaction tensors, as purely geometric functions, are independent of the multipole moments, and it is straightforward to differentiate them with respect to the atom positions (see SI for detailed expressions). The multipole response formally depends on both a Pulay term (response due to the basis center movement) and the 1-RDM response. However, the 1-RDM response is not explicitly needed if the wave function is at a variational stationary point, i.e., as long as the self-consistent HF/KS equations are converged. Given these considerations, the E^{II} gradient with respect to a QM atom j (denoted $\nabla_j E^{\text{II}}$) can be written as

$$\begin{aligned} \nabla_j E^{\text{II}} = & \sum_{\mu\nu} \gamma_{\mu\nu} (\delta_{\nu \in j} - \delta_{\mu \in j}) \times [v_{\mu}^{\text{II}} \langle \nabla | \mu \rangle \\ & + \sum_{\alpha} v_{\mu\alpha}^{\text{II}} \langle \nabla | (r_{\alpha} - R_{\mu\alpha}) | \mu \rangle + \sum_{\alpha\beta} v_{\mu\alpha\beta}^{\text{II}} \langle \nabla | (r_{\alpha} - R_{\mu\alpha}) \\ & (r_{\beta} - R_{\mu\beta}) | \mu \rangle] \end{aligned} \quad (13)$$

where on the right-hand side ∇ is the gradient with respect to the electron coordinate, and v_{μ}^{II} is the potential experienced by the atomic multipoles that the orbital μ is centered on. To derive the above expression, we have assumed that the 1-RDM is symmetric, and also used the $\nabla_j \phi_{\mu}(\mathbf{r}) = -\delta_{\mu \in j} \nabla \phi_{\mu}(\mathbf{r})$ property for an atom-centered orbital, and the identity $\langle \nabla | \nu | f(\mathbf{r}) | \mu \rangle + \langle \nu | \nabla f(\mathbf{r}) | \mu \rangle + \langle \nu | f(\mathbf{r}) | \nabla | \mu \rangle = 0$. Finally we note that the E^{II} gradient with respect to an MM atom only involves the gradient of multipole interaction tensors as there are no associated Pulay contributions.

Pseudobond Approach for the QM/MM Boundary.

When there are covalent bonds between some of the QM atoms and MM atoms, such as in the case of a QM treatment of amino acid side chains with an MM treatment of the backbone, dangling bonds arise in the QM subsystem from the QM/MM partitioning. Usually, the dangling bonds are either capped by an atom^{49–57} or by an orbital.^{58–61} We adopt the so-called pseudobond approach here because of the minimal changes it requires in a QM/MM implementation.^{50,52,57} The method introduces a specially parametrized cap atom to the dangling bond, placed at the location of the MM atom that is supposed to form the bond with QM atoms. For instance, in the above side chain-backbone partitioning, a special fluorine atom carrying special basis sets and a pseudopotential is positioned where the backbone α -carbon would be. Consequently, the C_{α} atom is treated as having dual QM and MM identities—it forms covalent bonds with the QM side chain via quantum mechanical interactions and also interacts with the MM backbone via a force field. The fluorine basis and pseudopotential are parametrized to mimic an sp^3 hybridized C_{ω} bonded to the side chain group $-R$. This is achieved by fitting the parameters to reproduce the geometry, charge distribution (excluding the methyl moiety), and relative protonation/phosphorylation energetics of a $R-CH_3$ molecule by a $R-F$ molecule. Herein, we use the same basis and pseudopotential forms as in refs 52, 57, and optimize for three DFT/basis combinations, $\omega B97X-V^{62}/6-31G^{**}$, $\omega B97X-V/6-311G^{**}$ and $\omega B97X-3c$.⁶³ The parametrization details and the resulting parameters are provided in the SI.

Implementation. We have implemented our QM/MM-Multipole method within the PySCF package taking advantage of GPU acceleration via the GPU4PySCF module. Given the full system atom positions, the QM atom element types, and MM charges (and optionally radii), PySCF is tasked with computing E_{QM} , $E_{\text{QM-MM}}$, and their nuclear gradients with respect to all atoms. The computationally expensive SCF calculations are offloaded to GPU routines in GPU4PySCF. The QM-MM electrostatic integral within region I (eq 7) is computed using the GPU implementation of the three-center-two-electron integrals in GPU4PySCF. In addition, the interaction tensors T , $\hat{T}$, and ψ , are computed on the GPU through CuPy⁶⁴ array operations, while the tensor contractions between the interaction tensors and multipoles are also performed by CuPy. In our QM/MM implementation, it is not necessary to employ a particle-mesh variant of Ewald,

which uses a real-space grid to interpolate the charges, to improve the computational scaling. This is because by choosing the Ewald real-space cutoff to be roughly the box length, the computational cost of both the real-space and k -space sums scales linearly with the number of MM atoms since we only compute the QM-QM and QM-MM Ewald energies in GPU4PySCF, and not the MM-MM Ewald energy. Aside from the overlap-like one-electron integrals, used for computing multipole moments and their gradients, which are evaluated on the CPU due to their relatively low cost, all QM/MM-related computational tasks are accelerated by the GPU.

We use LAMMPS⁶⁵ to compute E_{MM} and its nuclear gradients. This is straightforwardly achieved within LAMMPS input files, by turning off the bonded interactions within the QM subsystem and the electrostatics between the QM and MM atoms. In our current setup, GPU4PySCF and LAMMPS run concurrently in the background and communicate with an MD integrator, i-PI.⁶⁶ At each MD step, i-PI sends the box dimensions and atom positions via Internet or Unix sockets to both GPU4PySCF and LAMMPS and collects the computed energy and forces from them to integrate the nuclear dynamics. Such a communication scheme allows for the overlap of the LAMMPS CPU computation with GPU computation by GPU4PySCF. We note that LAMMPS itself can serve as the MD integrator as well if one desires to eliminate a dependency on i-PI. Presently since there is no communication between GPU4PySCF and LAMMPS, any choice of MD code can be easily adopted into this QM/MM communication scheme and requires no modification if it comes with an out-of-the-box i-PI interface, or if not, only small changes are needed to enable socket communication with i-PI.

■ COMPUTATIONAL DETAILS

Common Setup. We outline here computational details that the simulations below share in common. In all the simulations, we modeled the MM charges as Gaussian-distributed charges when computing the QM-MM interactions, with the exponent of the Gaussian on an atom given by the square of the inverse of its covalent radius in ref 67. The MM partial charges (used in $E_{\text{QM-MM}}$) and the Lennard-Jones parameters between the QM and MM atoms (used in E_{MM}) were kept the same as in the classical force field used in the simulation. We adopted an Ewald real-space cutoff roughly the size of the box length and estimated the Ewald parameters according to ref 68, given a desired 10^{-8} Hartree accuracy. The density fitting approximation to the electron repulsion integrals was employed, using the def2-SVP-JKFIT basis as the auxiliary fitting basis for the def2-SVPD basis, and the 6-311G**⁶⁹-RIFIT basis⁶⁹ for all the other atomic bases. For the pseudobond atoms, the fitting basis was generated by PySCF⁷⁰ using an even-tempered Gaussian expansion. The SCF convergence criteria were chosen to be 10^{-10} Hartrees for the energy and 10^{-6} Hartrees for the orbital gradient. In the MD simulations, we used the “time-reversible always stable predictor-corrector” method (ASPC)^{71,72} to predict the 1-RDM for the current MD step from the four preceding MD steps. The predicted 1-RDM was used as the initial guess for SCF and for dynamically determining region I based on a given error threshold (detailed further below). In all QM/MM NV T simulations, the temperature was controlled by a Langevin thermostat with a damping time constant of 100 fs. A time step of 0.5 fs was used for all QM/MM MD unless stated otherwise.

QM Water in MM Water. For this application, we considered three QM-water-in-MM-water systems, consisting of 7 QM water molecules embedded in respectively 1, 221, and 33234 SPC/Fw⁷³ MM water molecules. The initial configuration of the 221 and 33234 MM water systems was generated using Packmol⁷⁴ by placing 228/33241 water molecules in cubic boxes with 19.002 and 100.013 Å side lengths, according to a 0.99403 kg/L water density. The system was energy minimized and equilibrated for 50 ps at 298.15 K at the SPC/Fw MM level. The temperature was controlled by a Nose-Hoover chain and the dynamics were integrated using a 0.5 fs time step. The final equilibrated configuration was used for subsequent QM/MM simulations, where the 7 closest waters to the box center were selected as QM waters. All QM/MM calculations employed the PBE⁷⁵/def2-SVPD level of theory. The 7-QM-water-in-1-MM-water system was built by extracting the QM water configuration from the equilibrated 228-water structure and placing it at the center of a cubic box with a 27.399 Å side length. An MM water molecule was then positioned along one of the box diagonals with varying distances from the box center.

Guanosine Triphosphate Hydrolysis in Microtubules.

A model of guanosine triphosphate (GTP) hydrolysis in microtubules (MTs) was taken from a previous study⁷⁶ by Beckett et al. The system consists of a GTP-bound compacted interdimer of tubulin solvated in 56306 water molecules with 149 Na⁺ and 112 Cl[−] ions in a cubic box with a 121.5 Å side length. The previously equilibrated configuration was directly used as the starting point for our QM/MM simulation. The QM treatment was applied to the triphosphate part of the β -site GTP molecule, the coordinated Mg²⁺ ion, specific residue side chains from both the α and β subunits (α :Arg2, α :Glu254, α :Lys352, β :Asp67, β :Glu69, β :Asn99, and β :Thr143) and 19 solvation waters. The rest of the system was described by the CHARMM36m force field.⁷⁷ The pseudobond approach was used for the broken C $_{\alpha}$ –C $_{\beta}$ bonds of the amino acids and for the broken C4'–C5' bond for the GTP molecule. The backbone atoms of the QM residues and the H4', O4', C1', H1', C3', H3', O3', and H3T (CHARMM naming) atoms of the GTP molecule were set to have zero charge when computing the QM-MM coupling. All the MT simulations were performed in the NV T ensemble at 310 K, and the QM region was described at the PBE⁷⁵/6-311G**, B3LYP⁷⁸/6-311G*, ω B97X-V⁶²/6-311G**, or ω B97X-3c⁶³ levels of theory.

Chorismate Mutase. The chorismate mutase (CM) model we used is based on a previous study⁷⁹ by Ray et al. The system comprises a chorismate-bound enzyme solvated in 14821 water molecules with 12 Na⁺ ions in a box measuring 79.006 Å $\times$ 79.682 Å $\times$ 79.030 Å. We followed the same simulation setup as Ray et al., except for using our QM/MM-Multipole scheme and varying the QM region definition and DFT functional.

We considered two QM region definitions: one including the substrate and the catalytic residue Arg90 (S+R90), and another including the substrate with Arg90, Arg7, and Glu78 (S+R90+R7+E78). The pseudobond approach was used for the broken C $_{\alpha}$ –C $_{\beta}$ bonds of the QM residues, and the backbone atoms of the QM residues were set to have zero charge when computing the QM-MM coupling.

In the timing tests, the QM region was described by the PBE/6-311G**, ω B97X-V/6-311G**, or ω B97X-3c levels of theory. In the energy conservation test, we tested ω B97X-3c

with the S+R90 region. In the production runs for the catalytic kinetics, the QM region was described by PBE/6-31G**, ω B97X-3c, or a custom refined version of ω B97X-3c (see below for a detailed description of the customization).

We equilibrated the structure from Ray et al. using our QM(S+R90)/MM MD for 12 ps at 300 K, with a 1 fs time step, at both the PBE/6-31G** and ω B97X-3c levels of theory. The last frame of the PBE equilibration was used to initiate a 15 ps-long steered MD (SMD), biasing several interatomic distances to flip the Arg90 residue to form a hydrogen bond between the chorismate enolpyruvyl carboxylic oxygen and the arginine ϵ -nitrogen. The SMD was performed at the PBE/6-31G** level, and the detailed protocol can be found in the SI. The resulting structure with a flipped Arg90 forms two hydrogen bonds (HBs) with the chorismate while the structure taken from ref 79. features one chorismate-Arg90 hydrogen bond.

Additional QM/MM equilibration at both the PBE/6-31G** and ω B97X-3c levels was started from the last frame of the SMD, and run for 5 and 12 ps, respectively. The resulting structures were further equilibrated for 12 ps at the same level of theory with a larger QM region (S+R90+R7+E78). The ω B97X-3c equilibration was followed by an additional 12 ps of equilibration at the refined ω B97X-3c level.

For each combination of Arg90 conformation, QM region definition, and level of quantum theory, 11 configurations were selected from the last 10 ps of equilibration to initiate on-the-fly probability enhanced sampling (OPES) flooding simulations.⁸⁰ The configurations after the first 2 ps of each equilibration were used for initiating an SMD simulation to estimate the reaction free energy profile. The reaction coordinate (RC) ξ for both OPES flooding and SMD was the bond-making C1–C9 distance minus the bond-breaking C3–ester oxygen distance. A harmonic restraint with a 1000 kJ/mol/Å² force constant was used to drive the RC from its initial reactant value ($\xi \approx 1.8$ Å) to -0.9 Å over 7.5 ps of SMD. The free energy surface obtained from the SMD was used to determine the barrier height parameter and the bias-excluded region defined as $\xi < \xi^{\text{excl}}$ for the following OPES flooding. Detailed OPES flooding protocols are provided in the SI. All the enhanced sampling simulations (SMD and OPES flooding) used a 1 fs time step, with the enhanced sampling functionality provided by the PLUMED library.^{81,82}

We also performed geometry optimization and nudged elastic band (NEB) simulations to find reaction transition states starting from both the 1HB and 2HB Arg90 conformations. The first frames of the ω B97X-3c SMD with 1HB/2HB Arg90 and the S+R90 QM region were energy minimized at the same level of theory to produce local minimum energy reactant configurations. The last frames of the ω B97X-3c SMD were geometrically aligned to the reactant structures, and their product-like active-site conformations were combined with the geometries in the reactant state for the remaining part of the system. The combined conformations were energy minimized to find the product state minima. Eight images were interpolated between the resulting reactant and product states using the image dependent pair potential method.⁸³ NEB optimization was performed without a climbing image, targeting a maximum force of 0.15 eV/Å, and then with a climbing image using the same convergence criterion. The geometry minimization and NEB optimization were performed on the QM/MM PES with the S+R90 QM region and the ω B97X-3c functional, with the atoms of S

+R90+R7+E78 allowed to move. All energy minimization was performed using i-PI and all NEB calculations were carried out using the Atomic Simulation Environment (ASE).⁸⁴

We extracted the substrate and Arg90 geometries from the NEB conformations and performed single-point energy calculations in the gas phase. The Arg90 side chain was capped by a hydrogen positioned along the C α –C β direction at a distance of 1.08 Å from C β . The energies were computed using PBE/6-31G** and ω B97X-3c on every NEB image, and LNO–CCSD(T)⁸⁵ at five geometries: the reactant, the product, the transition state, and the two adjacent NEB images to the transition state. We performed the LNO–CCSD(T) calculations with cc-pVTZ and cc-pVQZ basis sets and used the two-point extrapolation formula⁸⁶ to extrapolate both the HF and correlation energies to the complete basis set (CBS) limit. All LNO calculations were performed with our recent implementation⁸⁷ in PySCFAD.⁸⁸ A natural occupation threshold of 10^{−5} was chosen for truncating the local virtual natural orbitals, while a natural occupation threshold of 10^{−4} was used for truncating the occupied natural orbitals. Such a choice of thresholds was found to give 0.05 kcal/mol mean absolute error in the LNO–CCSD(T)/cc-pVDZ energy on the five 1HB NEB geometries and 0.02 kcal/mol on the 2HB NEB geometries, compared to the canonical CCSD(T)/cc-pVDZ results. All the cc-pVDZ, cc-pVTZ, and cc-pVQZ calculations were also found to be converged to within 0.06 kcal/mol in MAE with the 10^{−4} occupied/10^{−5} virtual occupancy thresholds when comparing to the energies with larger thresholds (2 × 10^{−4} occupied/2 × 10^{−5} virtual). To reduce the error of the underlying quantum theory used in our MD simulations, we then refined the exchange parameters of the ω B97X-3c functional to reproduce the LNO–CCSD(T)/CBS reaction energy barriers in both the 1HB and 2HB Arg90 conformations, optimizing both the fraction of short-range and of long-range exchange to exactly reproduce the two reaction barriers. We then checked to make sure that the resulting refined ω B97X-3c functional also improved the energetics on the nontraining points as well. We observed more accurate energies and curvature around the transition state, noting that these are related to the probability of successful barrier crossing in classical rate theory.⁸⁹

RESULTS AND DISCUSSION

Error Control via Charge-Octupole Estimation. As discussed earlier, we truncate the multipole expansion at the quadrupole order (second order in the Taylor expansion) for the electrostatics between region II and the QM density in the reference cell, and thus the leading error in our approximation is the missing charge-octupole (the third order) interaction. This suggests that the interaction energy between the region II charges and the QM octupoles can serve as an error estimate,

$$\text{Err}_{\text{QM-MM}} \approx \sum_{i \in \text{MM}(\text{II})} \sum_{j \in \text{QM}(\text{I})} \sum_{\alpha\beta\gamma} \Omega_{i\alpha\beta\gamma} q_i T_{j\alpha\beta\gamma} \quad (14)$$

This estimate assumes that region I is sufficiently large that any higher-order terms (fourth order and above) are negligible. From here on, we consider region I to be defined by a sphere of radius R_{cut} centered at the QM subsystem geometric center. This cutoff truncates the short-range interactions of the QM octupoles and higher order multipoles with the charges outside region I, and should not be confused with the Ewald real-space cutoff.

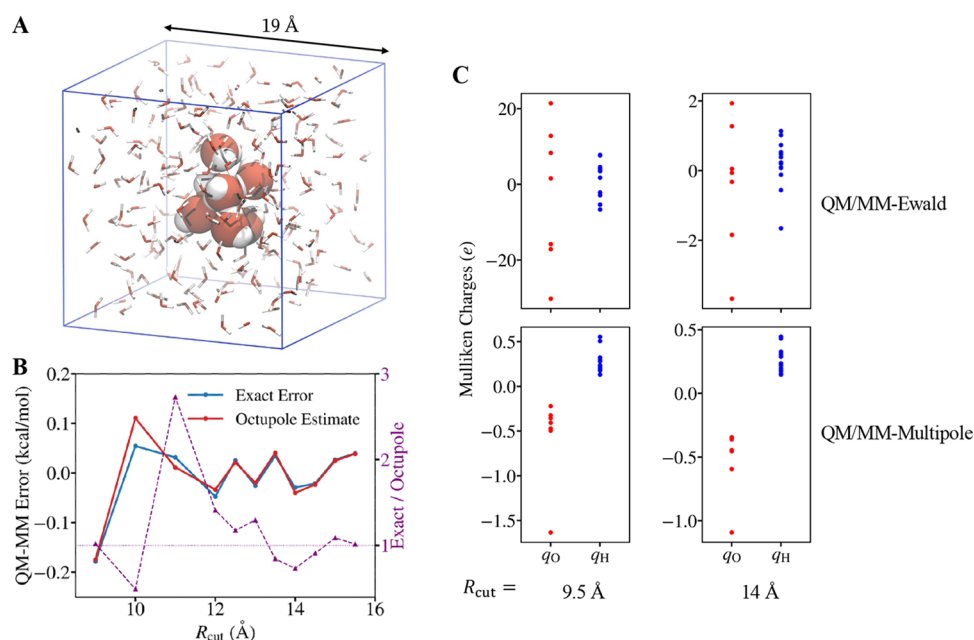


Figure 2. QM/MM analysis for a QM-water-in-MM-water system. (A) Configuration of the system where QM/MM calculations were performed. The QM water molecules are shown as spheres and the MM water molecules are shown as sticks. (B) Error of QM/MM-Multipole (up-to-quadrupoles) QM-MM energy as a function of the region I cutoff radius R_{cut} . (C) Mulliken charges of the QM atoms after up to 200 cycles of SCF with QM/MM-Ewald or QM/MM-Multipole using two R_{cut} values.

To validate the charge-octupole (octupole for short) error estimate, we examined a simple QM-water-in-MM-water system where computing the exact QM-MM electrostatic coupling energy is feasible. The system consists of 7 QM water molecules embedded in 221 MM water molecules (Figure 2A). The exact QM-MM electrostatic coupling was computed by the standard Ewald method, treating the electron density on the DFT grid as one group of point charges and the MM water molecules as another. The electron density used in the test was obtained from a converged QM/MM-Multipole SCF with $R_{\text{cut}} = 15.5$ Å. In Figure 2B, we show the exact error of including up to quadrupoles ($E_{\text{exact}} - E_{\text{approx}}$) versus the region II octupole error estimate as a function of R_{cut} . (We did not reoptimize the density with different R_{cut} in this test, thus the R_{cut} dependency is entirely due to the varying number of exactly computed QM-MM interactions). We find a good agreement between the octupole estimate and the exact error beyond $R_{\text{cut}} \approx 12.5$ Å. For a smaller R_{cut} , the octupole error estimate remains qualitatively correct and is of the same order of magnitude as the exact error.

We next examine a more complex system, a model of GTP hydrolysis in microtubules. The system composition and QM subsystem definition are detailed in the Computational Details section as well as shown in Figure 3A,B. The electronic structure was described at the ω B97X-3c level in this analysis. We estimated the exact error by comparing the energy at $R_{\text{cut}} = 40$ Å (which we take as exact) with the SCF-converged energy $E(R_{\text{cut}})$ at various R_{cut} values. Unlike in the previous example, this error depends on R_{cut} not only through the omission of octupole and higher interactions with region II but also through the R_{cut} -dependency in the converged QM density, from the absence of polarization effects at the above-quadrupole level by region II charges. According to the linear response theory of HF/KS,⁹⁰ the error in the density is proportional to the error in the electronic Hamiltonian. As such, the lack of octupole polarization by a charge at distance R

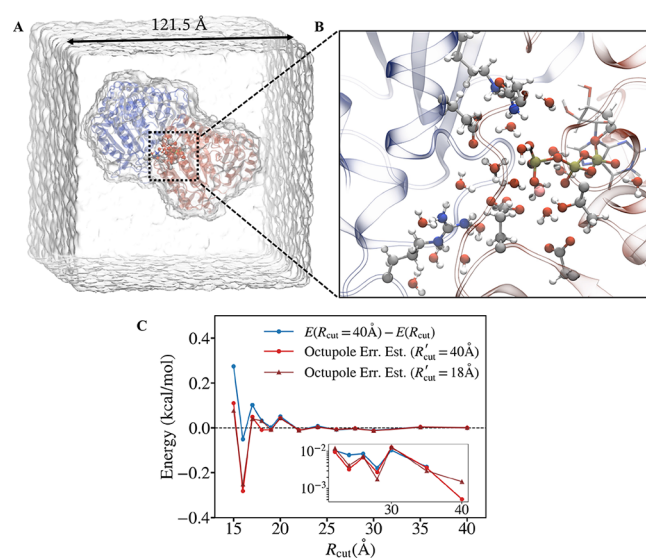


Figure 3. QM/MM analysis of a model for microtubule-mediated GTP hydrolysis. (A) System configuration used for the QM/MM calculations. The QM atoms are shown as spheres, the protein is depicted as ribbons (with subunit α in blue and subunit β in red), and the rest of the system is represented by transparent surfaces. (B) Zoomed-in view of the QM region. The QM atoms are shown as balls and sticks, the MM part of GTP is represented by thin sticks, and the protein is shown in transparent blue (α subunit) and red (β subunit). The atoms are colored as follows: hydrogen (white), oxygen (red), nitrogen (blue), carbon (gray), magnesium (pink), and phosphorus (gold). (C) Error in the QM/MM-Multipole SCF-converged energy as a function of the region I cutoff radius R_{cut} , assuming the energy obtained with $R_{\text{cut}} = 40$ Å is the ground truth. The octupole error estimate was computed using the density converged with a cutoff of R'_{cut} .

introduces a density error that scales in the same way as the charge-octupole interaction ($\propto 1/R^4$). Integrating over the

Table 1. Computational Performance of the QM/MM MD Simulations Using 32 CPU Cores (AMD EPYC 7742) and One A100 GPU^a

system	PBE/6-311G**	B3LYP/6-311G*	ω B97X-3c	ω B97X-V/6-311G**
MT	1900 $\pm$ 90	1510 $\pm$ 70	730 $\pm$ 20	370 $\pm$ 20
CM (S+R90)	11200 $\pm$ 600	9900 $\pm$ 500	6600 $\pm$ 300	4400 $\pm$ 200
CM (S+R90+R7+E78)	6900 $\pm$ 300	5800 $\pm$ 300	3290 $\pm$ 80	1790 $\pm$ 80

^aThe reported values are the mean number of steps per day, with the fluctuations computed from the standard deviation over 100 MD steps.

whole region II, this gives an accumulated error scaling as $\int_{R_{\text{cut}}}^{\infty} 4\pi R^2 / R^4 dR \propto 1/R_{\text{cut}}$. Due to the variational condition, the total energy error scales quadratically with the density error, yielding a $1/R_{\text{cut}}^2$ decay, faster than the $1/R_{\text{cut}}$ scaling of the total octupole interaction with region II. Hence, the leading term of the error at large R_{cut} is still the missing octupole interaction, the same as in the previous test with a frozen density. As shown in Figure 3C, the octupole error estimate obtained by simply summing the missing charge-octupole interactions agrees well with the “exact” error at sufficiently large R_{cut} . Furthermore, since the energy error due to the density error decays quickly ($1/R_{\text{cut}}^2$), the octupole error estimate may be computed using a density converged with a small region I (defined by R'_{cut}), and we show this is indeed the case in Figure 3.

We now discuss the practical use of such an octupole error estimate to determine R_{cut} in production simulations. At the first MD step, one may perform a QM/MM-Multipole SCF calculation with an empirically chosen R_{cut} to generate a QM density guess. A new R_{cut} can be determined by incrementing R_{cut} from a chosen lower bound until the octupole error estimate computed with the density guess falls below a desired accuracy ϵ . If the new R_{cut} is larger than the initially guessed one, one may either rerun the first MD step with the determined R_{cut} or simply discard this step from the statistics. In the following MD steps, a proper R_{cut} can be estimated on the fly using the ASPC-predicted density from previous steps.

Timings. We evaluated the computational efficiency of our QM/MM-Multipole approach on two biomolecular systems: a chorismate mutase model and a model for microtubule-mediated GTP hydrolysis. We note that many aspects of these calculations (ranging from the auxiliary basis, here chosen as 6-311G-RIFIT in all cases below, to the detailed implementation) can be further tuned to improve the efficiency. Thus, the timings reported below should be viewed only as an initial guide to the performance.

For the PBE and B3LYP calculations, the pseudobond parameters for ω B97X-V were used to avoid building another parametrization. The wall time per MD step was averaged over 100 MD steps, with a fixed $R_{\text{cut}} = 25$ Å. The MT system contains 156 QM atoms, and the CM system has 43 QM atoms in the S+R90 definition of the QM subsystem and 72 QM atoms in the S+R90+R7+E78 definition of the QM subsystem. The number of computational basis functions (N_{AO}) is 1756, 532, and 846 using the 6-311G** basis in the MT and for the two CM QM subsystem definitions, respectively, and $N_{\text{AO}} = 1513$, 475, and 744 for the 6-311G* basis set. The ω B97X-3c functional comes with a specially parametrized double- ζ basis, and $N_{\text{AO}} = 1309$, 398, and 637 in the MT and two CM QM subsystem definitions. The timing results are provided in Table 1.

We observe that in our current implementation of periodic ab initio QM/MM MD, simulations with a hybrid density

functional (ω B97X-3c) on systems with >100 atoms (>1000 atomic basis functions) can be run with a throughput of $\sim$ 1000 MD steps per day on a single GPU. Although a careful comparison will require controlling many other details and will be considered in a later study, the speed of B3LYP/6-311G* on the MT system appears to be roughly comparable to that reported for B3LYP on a system with a similar size in recent GPU implementations of nonperiodic ab initio QM/MM MD.⁹¹

SCF Stability. If one ignores the dipole and quadrupole terms, the QM/MM-Multipole approach reduces to the QM/MM-Ewald approach using Mulliken charges.²³ (More precisely, the original QM/MM-Ewald approach additionally fixes region I to be the unit cell but we allow a flexible choice of region I herein). It is known that the QM/MM-Ewald method with Mulliken charges suffers from instabilities in the SCF optimization when diffuse basis functions are used. The instability is related to the definition of the Mulliken charges, which becomes increasingly unsatisfactory in diffuse bases (although a description of the electrostatic potential at long distances is still provided by the full set of Mulliken multipoles). In the test system shown in Figure 2A, we show that we could not converge the SCF within 200 cycles using the QM/MM-Ewald approach even when R_{cut} was set to half the box length. After extending region I beyond the unit cell, with $R_{\text{cut}} = 14$ Å, we were able to finally converge the QM/MM-Ewald SCF. The converged electron density, however, still has large errors revealed by large positive/negative Mulliken charges on the oxygen/hydrogen atoms. In contrast, the QM/MM-Multipole approach is always stable and produces reasonable Mulliken charges, even with the smaller R_{cut} .

Energy Conservation. Since we treat the electrostatics in regions I and II differently, the potential energy surface of our QM/MM-Multipole approach is not strictly smooth. In Figure 4A, we check the smoothness of the energy surface when one MM water is moved from region I to region II with a fixed $R_{\text{cut}} = 17$ Å. In contrast to the large energy jumps observed for QM/MM-Ewald, we see smooth (on the scale of the plot) energies in the QM/MM-Multipole approach. The large jump of the QM/MM-Ewald energy right at $R = R_{\text{cut}}$ occurs when the boundary cuts through the MM water molecule. This is because although the leading electrostatic interaction with a water molecule should only involve the water dipole, when the water molecule is cut, the electrostatics in QM/MM Ewald is separated into interactions with individual charges on the water molecule, some of which are treated exactly and some of which are treated in the QM monopole approximation. There is thus a large residual charge interaction term which is unphysical. The QM/MM Ewald energy still has a noticeable jump, however, even when we compare energies just before and after a whole water molecule is moved to region II, when this charge interaction error is absent.

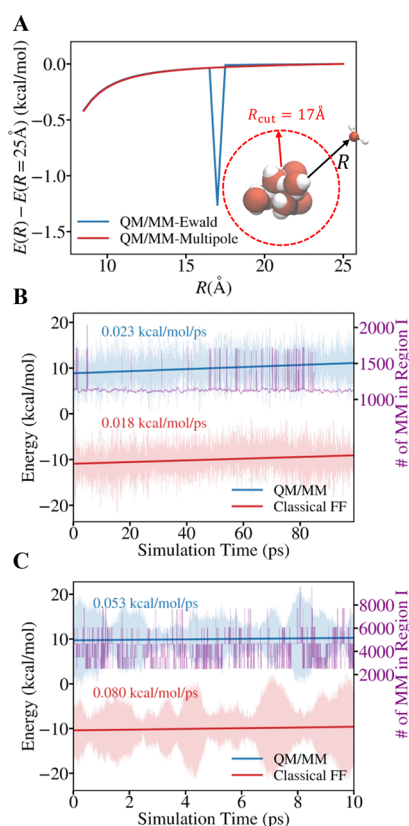


Figure 4. Energy conservation of the QM/MM-Multipole method. (A) Potential energy surface with a constant region I R_{cut} as a function of an MM water (shown as balls and sticks) distance from the QM cluster (shown as spheres) center. (B) Energy conservation in a QM-water-in-MM-water system. Linear fits to the energy time series are plotted as solid lines, and the text in the plots shows the slopes of the fits. (C) Energy conservation in a chorismate mutase model.

We next examine whether the QM/MM-Multipole PES is smooth enough not to lead to a significant energy drift during long MD runs. We first tested this on a system consisting of 7 QM water molecules and 33234 MM water molecules. A harmonic potential with a force constant of 0.2 kcal/mol/Å² was applied to the oxygen atoms of the QM waters to restrain their relative distances from the geometric center of the QM oxygens. The initial R_{cut} was set to 16 Å, and the subsequent R_{cut} was determined on the fly every 10 MD steps by searching starting from 13 Å with a step-size of 1 Å until two consecutive R_{cut} values both give an octupole error estimate under $\epsilon = 2 \times 10^{-4}$ Hartree. NV E dynamics was run for 99 ps after 1 ps of NV T equilibration at 310 K. Figure 4B shows that the energy conservation of the QM/MM-Multipole dynamics is of a similar quality to that of classical force field dynamics, despite fluctuations in the number of MM atoms in Region I arising from the on-the-fly selection at every MD step of the regions using an R_{cut} that varies with time.

We further examine the energy conservation in the CM system as a more complicated example. NV E dynamics was run for 10 ps after 250 fs NV T equilibration under 300 K. The R_{cut} was dynamically determined every 5 MD steps by searching from 16 Å with a step-size of 2 Å until two consecutive R_{cut} values both give an error estimate under $\epsilon = 2 \times 10^{-5}$ Hartree. The more frequent R_{cut} search and the tighter ϵ result in a larger fluctuation in the number of region I atoms.

As shown in Figure 4C, the energy conservation of our QM/MM dynamics is again comparable to that of classical MD.

Example Application: Catalytic Kinetics in Chorismate Mutase. Chorismate Mutase (CM) is a critical component in the primary and secondary metabolism of fungi, bacteria, and plants.⁹⁵ It catalyzes the Claisen rearrangement that transforms chorismate into prephenate (Figure 5A), an important precursor for the biosynthesis of aromatic amino acids.^{95–97} The catalytic kinetics of this reaction were recently studied by Ray et al. via MD enhanced sampling simulations on a QM(PBE)/MM PES.⁷⁹ They computed the reaction rate using a variant of the flooding method⁹⁸ called OPES flooding,⁸⁰ which adaptively adds bias potentials to the reactant state to accelerate barrier crossings that would otherwise be inaccessible to unbiased simulations. The computed QM/MM reaction rate constant was found to be somewhat higher than the experimentally measured catalytic rate constant, and this discrepancy was attributed by the authors to the neglect of the free energy cost for the substrate to adopt a “reactive” conformation.

We now re-examine this reaction, following the simulation setup of Ray et al. as closely as possible, but further explore how the theoretical kinetic predictions depend on the MD sampling, the QM region definition, and the employed QM theory. In all our simulations, we used the same scheme to determine R_{cut} as in the energy conservation test. We verified our QM/MM scheme and implementation by reproducing the QM(PBE/DZVP)/MM result of CP2K by Ray et al. using our QM(PBE/6-31G**)/MM-Multipole MD. We found that the calculated rate constant ($5.1 \times 10^3 \text{ s}^{-1}$) indeed overestimates the experimental one of 16 s^{-1} while being of the same order of magnitude as the value reported by Ray et al. (9433 s^{-1}).

We first consider how the rate constant depends on the level of quantum theory. To do so we used two more accurate hybrid density functionals: ω B97X-3c and a refined version of it parametrized to reproduce the LNO-CCSD(T)/CBS reaction barriers for this system (see Computational Details). We kept the QM region definition unchanged and started from the same structure as Ray et al. (with appropriate equilibration using the corresponding functional). By changing the quantum description, we arrive at an opposite conclusion—the QM/MM reaction rate constant now significantly underestimates the experimental catalytic rate constant (the first row of Table 2). The qualitatively different conclusion is a result of the inaccuracy in the pure functional treatment of the QM region, with its substantial underestimate of the reaction barrier, as shown by the gas-phase energetic analysis (Figure 5H). Assuming that the estimated experimental rate constant is accurate and the refined ω B97X-3c functional PES is almost exact, the deviation of the computed rate constant must be a result of either insufficient MD sampling (statistical error) or an inadequate QM region size (a source of systematic error).

As shown in Table 2, our calculated rate constant does not appear to suffer from a large statistical error, but in the simulations initiated from the conformation of ref 79, we never observed a double hydrogen bond between the catalytic Arg90 and the substrate, even though this is a known binding mode found in a CM crystal structure (PDB id: 1COM).⁹⁹ Instead, Arg90 forms only one hydrogen bond with the substrate (Figure 5B), resembling the conformation captured in another crystal structure (PDB id: 2CHT)¹⁰⁰ that was used to initiate the simulations of Ray et al. The main difference between the two binding modes is in the orientations of the ϵ -nitrogen/

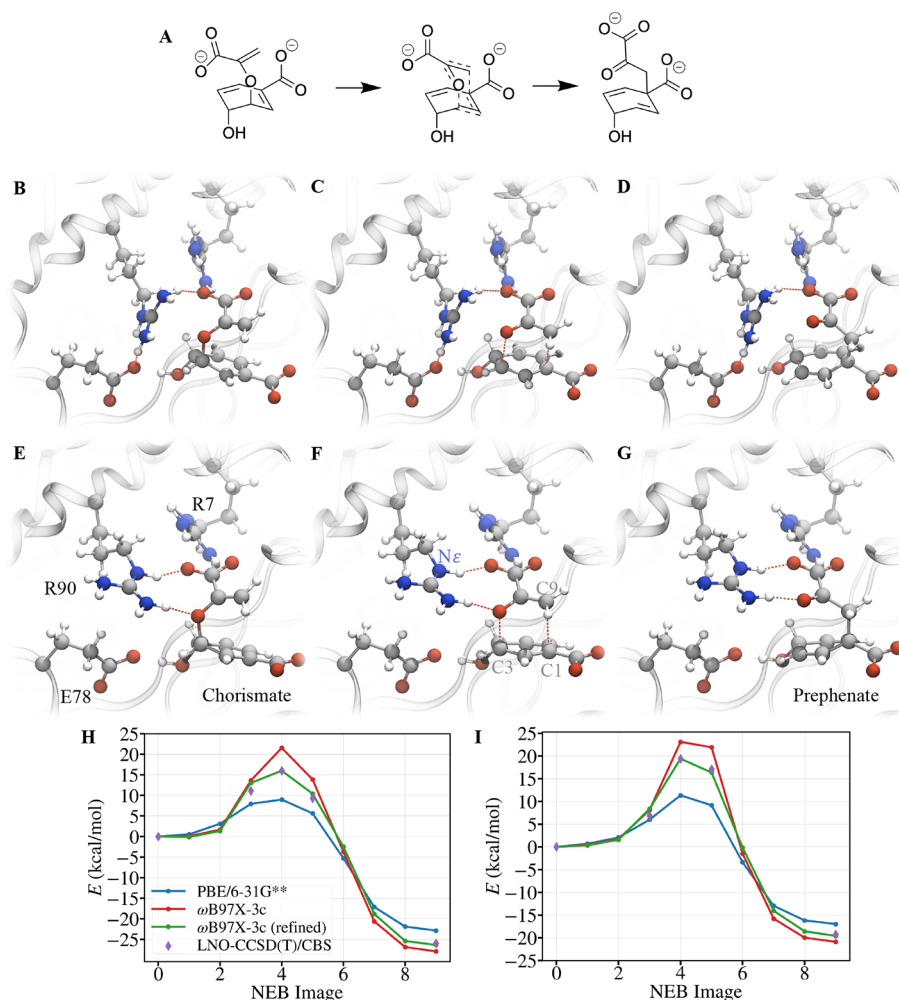


Figure 5. (A) Chorismate mutase catalyzed reaction. (B–D) Nudged elastic band optimized geometries of the reactant, the transition state, and the product when Arg90 forms a single hydrogen bond with the substrate. (E–G) Optimized reactant, transition state, and product geometries when Arg90 forms double hydrogen bonds with the substrate. Energies are aligned at the reactant state. (H) Gas-phase energies on NEB geometries for the 1HB conformation. (I) Gas-phase energies on NEB geometries for the 2HB conformation.

Table 2. Catalytic Rate Constant of Chorismate Mutase from *Bacillus subtilis* in s^{-1} ^a

expt.	binding mode/QM region	theory		
		PBE/6-31G**	ω B97X-3c	ω B97X-3c (refined)
16 ± 14 ^b	1HB/S+R90	$(5.1 \pm 0.3) \times 10^3$	$(2.7 \pm 0.4) \times 10^{-8}$	$(4.0 \pm 0.7) \times 10^{-3}$
	2HB/S+R90	$(9.2 \pm 0.5) \times 10^5$	$(1.1 \pm 0.2) \times 10^{-4}$	$(4 \pm 2) \times 10^{-2}$
	2HB/S+R90+R7+E78	$(1.2 \pm 0.2) \times 10^7$	$(5.8 \pm 0.7) \times 10^{-4}$	1.1 ± 0.2

^aTheoretical rate constant errors are reported as the square root of the rate constant estimator variance. Statistical analysis details are provided in the SI. ^bThe experimental catalytic rate constant at 300 K was computed from the fitted temperature-dependent rate constant $k_{\text{cat}} = k_B T/h \exp(-\Delta H/k_B T + \Delta S/k_B)$ with parameters $\Delta H = 12.7 \pm 0.4$ kcal/mol and $\Delta S = -9.1 \pm 1.2$ cal/mol/K, taken from ref 92, divided by the solvent kinetic isotope effect (KIE) measured at 303.15 K.⁹³ The error in this rate constant was estimated by the uncertainty propagation equation⁹⁴ by assuming the errors of ΔH and ΔS are independent and no error in KIE. The normalization by KIE is for direct comparison to the theoretical rate constants obtained with classical nuclear dynamics.

hydrogen of Arg90. In 2CHT, an HB between the Arg90 terminal nitrogen and the substrate carboxylic oxygen is found with no other HB interactions between the two moieties. In 1COM, two HBs are found, one between the ϵ -nitrogen and the chorismate carboxylic oxygen, and another between the terminal nitrogen and the chorismate ester oxygen. The transition from the 1HB conformation to the 2HB conformation requires a flip of Arg90, which is not accessible in our QM/MM equilibration.

To investigate how the Arg90 conformation affects the reaction rate, we added biasing potentials to artificially accelerate this flip and then equilibrated the resulting 2HB structure after removing the bias (see Figure 5E for an example of the 2HB structures). We performed OPES flooding on this new structure to compute the rate constant. We found then that the calculated rate constants were enhanced regardless of the functional choice, with the ω B97X-3c one even increasing by 4 orders of magnitude (the second row of Table 2).

The sensitivity of the rate to the local protein conformation underlines the challenges of converging the sampling, especially when one has no prior knowledge of the existence of alternative binding modes. Interestingly, the gas-phase single point energetics show a lower reaction barrier for the 1HB binding mode than for the 2HB binding mode (Figure S4,I), contradicting the faster reaction rate constants obtained from OPES flooding simulations with the 2HB conformation. This illustrates the importance of entropic contributions and the appropriate modeling of the chemical environment.

It has been conjectured that the catalytic mechanism of CM involves an acid/base attack step due to the considerable measured solvent kinetic isotope effects (KIE)⁹³ on the apparent catalytic rate constant, $k_{\text{H}_2\text{O}}/k_{\text{D}_2\text{O}} = 2.23$. The sensitivity of the theoretical rate constant on the hydrogen-bonding pattern suggests, however, that the solvent KIE results from the hydrogen bond strength's dependence on the quantum nature of the protons; further simulations with an explicit treatment of nuclear quantum effects (NQEs) should be carried out to explore this point.

We also tested the sensitivity of the rate constant to the definition of the QM region by further including Arg7 and Glu78 in the QM region. When switching these residues from an MM to a QM description, we found all the calculated rate constants further increase (the third row of Table 2), consistent with substantially lower enzyme activities in mutagenesis analyses of Arg7 and Glu78.^{101,102} The QM treatment allows a proton on Arg90 to be shared with Glu78 in PBE/6-31G** simulations, and even occasionally, for a proton transfer (PT) from Arg90 to Glu78 (1 in 11 simulations of $\omega\text{B97X-3c}$ and 4 in 11 simulations of the refined $\omega\text{B97X-3c}$). However, the difference in the lnk of the refined $\omega\text{B97X-3c}$ with 2HB/S+R90+R7+E78 is not statistically significant between the runs where PT happens and the ones where it does not (the *p*-value from a Welch's *t*-test is 0.36 assuming equal means as the null hypothesis). This lack of statistical significance suggests that the enhanced rate cannot be directly attributed to the occurrence of PT alone. Instead, the observed rate increase may be explained by other factors, including the potential charge transfer and the differences in the electrostatic representation between QM and classical MM charges.

Our final theoretical rate constant with the refined $\omega\text{B97X-3c}$ functional, the (larger) S+R90+R7+E78 QM region definition, and the double hydrogen-bonded Arg90 conformation, shows a reasonable agreement with the estimated experimental one. At the same time, our simulations illustrate the difficulties of obtaining a fully converged result. Improving this agreement will require improving the errors in the theoretical treatment, from the imperfect electronic structure, size of QM region, sampling, nuclear quantum effects, and probably the rate theory as well (the correct estimate of rates depends on a careful choice of OPES flooding parameters according to ref 80). Within DFT treatments, the quality of the functional remains perhaps the largest source of uncertainty, with almost 11 orders of magnitude difference in rate constant between different functionals.

From the PBE data, where the rate constant with the 2HB Arg90 conformation and the larger QM region is much faster than the experimental value, one might conclude that the Claisen rearrangement step is not rate-limiting for the whole catalytic cycle, and perhaps the substrate leaving step is instead rate-limiting, as has been hypothesized in some earlier

experiments.⁹² However, our results using the improved PES provided by the refined $\omega\text{B97X-3c}$ functional suggest the chemical transformation step is rate-limiting regardless of the Arg90 conformation or the QM region definition. This aligns with the proposal that electrostatic transition state stabilization is fundamental for CM catalysis,¹⁰³ and consistent with data that finds impaired enzyme activity after mutating Arg90 into a neutral citrulline, an arginine analog that is still able to preorganize the chorismate into its "reactive" conformation.

CONCLUSIONS

In this work, we described a QM/MM implementation with an improved treatment of the periodic QM/MM electrostatics which leverages recent advances in GPU DFT algorithms to enable the practical use of advanced hybrid density functionals. Our benchmarks verify its numerical stability and illustrate the possibility of controlling and converging the error associated with the electrostatics.

While this work has not focused on achieving a computationally optimal implementation, the open-source availability of our code and the already competitive level of performance achieved form a foundation for broader community efforts to build an even faster QM/MM code.

Our QM/MM investigation of the chorismate mutase reaction highlights the critical importance of converged conformational sampling and an accurate QM treatment in theoretical kinetic studies of enzymatic reactions. Despite the simplicity of this system, it appears challenging to obtain a fully converged theoretical description. We view this work as a first step toward improving the quality of sampling and QM modeling as part of our ongoing efforts to obtain a precise understanding of enzymatic reactions from direct quantum mechanical simulations.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.4c01698>.

Details on multipolar Ewald and pseudobond parametrization, additional computational details on chorismate mutase, rate calculations, and statistical error analysis (PDF)

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Author Contributions

C.L. and G.K.C. formulated the project. C.L. carried out the implementation and calculations. C.L. and G.K.C. discussed the results and interpretation and wrote the manuscript.

Notes

The authors declare no competing financial interest.

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